C6 — The strategy zoo: five published market makers, and RL on all five

Lab notebook: `experiments/c6_strategy_zoo/README.md`.

> **At a glance**
>
> - Five published market makers on nine assets, each with its published spread, a per-asset widened spread, and TD3 on four multipliers on top. Held out: 11 sessions on ETHUSDC and XRPUSDC, 3 on the other seven.
> - **Widening is the largest effect in Part C**: a median +2.25 points of cumulative return, better in 25 of 33 (asset, strategy) pairs. The floor an asset needs is 9–19× its trade-intensity decay $1/k$ on eight of nine.
> - **RL on top does not beat the widened baselines.** With five seeds, recipe v1 loses to its baseline in 8 of 8 pairs. Recipe v2 — a residual policy started on the baseline, five-second decisions, a partly counterfactual reward — beats v1 in 8 of 8 but its baseline in only 3 of 8, and the seed spread is larger than any gain.
> - More orders in the grid does nothing; the policies do not learn to respond to the alpha signal.

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1. The task

Not a ladder phase. This one comes from a direct request after C3 landed:

> try to use more days for training and held-out and larger spread (dependent on asset) for baseline and then RL mod (also try more orders in grid) — build baseline and RL modification for all these strats — prepare master table for everything (including trade intensity) and pnl + inventory plots

with five hftbacktest example notebooks attached. Five things, then:

1. more days, on both sides of the split;
2. a wider, asset-dependent spread for the baselines, and then the RL modification on top of that;
3. deeper grids;
4. a baseline **and** an RL modification for each of the five strategies;
5. one master table covering everything, trade intensity included, plus PnL and inventory plots.

It also happens to answer the two items C5 has been carrying (`ladder.md`) — fold the basis and APT alpha signals into the state and re-run C3, and check whether the learned coefficients respond to the alpha signal rather than needing a hand-written extension — so those are resolved here rather than separately.

2. The goal, and the objection it answers

C3's result was defensible but narrow. On ETHUSDC over five held-out sessions no learned configuration beat the fixed GLFT formula in absolute terms, though all of them lost less: the best TD3 policy returned −0.63% of book against the baseline's −8.44%.

The objection writes itself. **GLFT was the wrong baseline.** C1 showed it loses to adverse selection on this venue — it quotes about 1.7 bp out on an asset whose fills cost more than that — so "RL loses less than a losing strategy" is a weak claim, and a reader is entitled to suspect that a profitable baseline would leave RL nothing to add.

C6 takes that objection seriously in three ways at once. It runs the four other strategies hftbacktest publishes alongside GLFT, so the answer is about the method rather than one model. It widens the spread until the baselines are not obviously broken, so RL is asked to improve something that works. And it does both on twice as many training days and more than twice as many held-out ones.

3. The five strategies, in one set of units

3.1 What they share

Every one of them is a grid quoter. At each 100 ms iteration it forms a fair price p^*, a half spread h and an inventory skew s, sets a reservation price

$$p^r = p^* - s \cdot \frac{q}{Q}\,m$$

where q is the position in units of the asset, m the mid and Q the notional position cap (so the second factor is the normalised inventory, -1 at a full short and $+1$ at a full long), and then lays `grid_num` orders a side at

$$p_i^{\text{bid}} = p^r - h - i\,g, \qquad p_i^{\text{ask}} = p^r + h + i\,g$$

aligned to a grid of step g, with any order already correctly placed left alone so it keeps its queue position. What differs between the five is only p^*, h and s.

3.2 Where each one gets its numbers

`glft` — Guéant, Lehalle & Fernandez-Tapia (paper), equations (4.6)/(4.7), with $\Delta = 1$ and $\xi = \gamma$. Fair price is the mid; the half spread and skew come from a live fit of the order-arrival intensity $\lambda(\delta) = Ae^{-k\delta}$ and the mid-price volatility σ:

$$h = \left(c_1 + \tfrac{\Delta}{2}\,\sigma c_2\right)\text{adj}_1, \qquad s = \sigma c_2\,\text{adj}_2$$

$$c_1 = \frac{1}{\xi\Delta}\log\left(1 + \frac{\xi\Delta}{k}\right), \qquad c_2 = \sqrt{\frac{\gamma}{2A\Delta k}\left(1 + \frac{\xi\Delta}{k}\right)^{\frac{k}{\xi\Delta}+1}}$$

with $\gamma = 0.05$, $\text{adj}_1 = 1$, $\text{adj}_2 = 0.05$ — the notebook's values, and C1's `glft-grid`. C1's spread-floor result applies here: $\inf_{\gamma>0} h(\gamma) = 1/k$, so no amount of risk aversion lets this quoter sit inside a depth of $1/k$ from the mid. (The notebook fits k per *half*-tick; the tables convert it to ticks and bp.)

`hfgrid` — no model of arrival intensity at all. The fair price is the micro-price and the half spread is a multiple of realised volatility, $h = 5\sigma$, with a multiplicative skew. Note that at the touch the micro-price carries exactly the information the queue imbalance does:

$$\text{micro} - \text{mid} = \frac{P_a - P_b}{2} \cdot \frac{q_b - q_a}{q_a + q_b} = \frac{\text{spread}}{2} \cdot \text{imbalance}$$

identically — C2 measured this rather than assuming it — so `hfgrid` and `obi` differ in how far into the book they look, not in the kind of signal.

`apt` — a lead-lag fair price, $p^* = (1 + \beta r_{\text{ref}})\,p_{t-5\text{m}}$, where r_{ref} is the reference instrument's five-minute return and $p_{t-5\text{m}}$ our own mid five minutes ago. Half spread and skew are both 3 bp.

`basis` — $p^* = p_{\text{ref}} + \overline{(p - p_{\text{ref}})}$, the reference price plus a five-minute rolling mean of the basis. Same 3 bp spread and skew.

`obi` — $p^* = m + c_1\alpha$ with α the order-book imbalance over a band $\pm 0.1\%$ wide, standardised over a ten-minute rolling window.

3.3 Everything in basis points

The notebooks state their parameters in ticks or in price units of whichever asset they were written for, and those numbers do not travel. `half_spread = 80` is 30 bp on 2023 BTC at $27,000 and meaningless on XRP at $3. So C6 converts once, in `strategies.py`, and every spread, skew and grid step is a ratio of price.

The conversion is also the one place a transcription error hides silently — a wrong γ produces an obviously broken run, a wrong unit produces a plausible one — so it is checked rather than trusted. For the skew, C1 shifts by $s_{\text{tick}} \cdot q$ ticks with q in units of the asset; C6 shifts by $s_{\text{px}} \cdot qm/Q$ in price with $s_{\text{px}} = s_{\text{tick}}\,\tau\,Q/m$. The mid and the cap cancel and the two are the same shift; `run_checks.py` confirms it to 5e-16 over 2,000 random states.

The OBI conversion doubles as an independent check on itself, because the notebook gives two tunings per asset and they must agree once converted:

tuning	asset	published	converted
wide	BTC ≈ 27,000	half 80, skew 3.5/lot, c_1 160	29.6 bp, 25.9 bp, 59.3 bp
wide	ETH ≈ 1,800	half 5, skew 0.2/lot, c_1 10	27.8 bp, 22.2 bp, 55.6 bp
volume	BTC ≈ 27,000	half 10, skew 2/lot, c_1 20	3.7 bp, 14.8 bp, 7.4 bp

They do. C6 takes the volume tuning as the `obi` baseline: it reports the higher Sharpe in the notebook (14.01 against 10.83), and a tuning that barely trades is not a market maker to compare market makers against.

Note what the wide tunings say on their own, before any of C6's results: two of the five published strategies chose a half spread near **30 bp**, roughly twenty times what C1's calibrated GLFT quotes. That is the prior behind item 2 of the request, and it was in the notebooks all along.

4. The data

4.1 Widening the split, not re-picking it

The split was fixed once, in C1, and C3's held-out claims rest on it. Re-picking it now — after seeing that C1's baseline loses — would be the exact failure C4's third item exists to catch, and doing it deliberately would be worse than doing it by accident.

So it is widened. `TRAIN_DAYS_EXT` and `TEST_DAYS_EXT` are supersets of C1's five-day lists on the same side of the same `SPLIT_DATE = 2025-09-15`:

- **train** — ten of the twenty days available before the split, spread across the window rather than clustered at its start: 08-26, 08-27, 08-28, 08-29, 09-01, 09-03, 09-05, 09-08, 09-10, 09-12.

- **held out** — **every** day from the split to the end of the data, eleven of them, with no selection at all. A held-out set with nothing chosen about it is the only one that cannot be accused of cherry-picking.

Nothing C1, C2 or C3 reported moves, because their days are a subset of these and each session is replayed independently.

The XRP pair started with four training and four held-out sessions, because two symbols per day is roughly three times the bytes of one. Four held-out sessions proved too few once the seed spread was measured (§8.4), so the pair's held-out set was widened the same way, to all eleven days from the split onward. The other seven assets keep three and three: they are there for breadth, and the session count is printed next to every row so the two tiers are never read as the same kind of evidence.

4.2 What tier the data is, and what that costs

Market-by-price (L2), not market-by-order (L3), verified from the files rather than from the dataset description: `order_id` is zero on all 16,885,548 rows of a day, and the event types are 16.5M incremental depth updates, 394k trades, six depth-clears and 6,000 snapshot rows -- the snapshot and clear being why each day is self-contained and replayable on its own.

The consequence is the one that matters most for every number in this phase. **With no order identities, queue position cannot be observed and has to be modelled.** hftbacktest guesses where a resting order sits within its price level; C6 uses `power_prob_queue_model(2.0)` throughout, and C1 measured that this choice moves results more than anything the strategy does.

It is also the reason §5 ends where it does. The tape says a quote breaks even at almost any depth; the simulator says C1's baseline lost 0.80 bp per fill. The gap is *which* aggressive trades reach a resting order, and that is exactly what L3 would answer and L2 cannot. Every spread floor in §8.1 is therefore conditional on the queue model, and tighter quotes are where that conditionality bites hardest -- a quote at the touch is almost all queue position, a quote 30 bp out is almost none.

Order latency is synthesised for the same reason: the dataset carries no order-latency feed, so entry and response latency are 4x and 3x the observed *feed* latency (`local_ts - exch_ts`), following hftbacktest's own tutorial, which is about a 20 ms round trip on these days.

4.3 The substitution APT and basis need

Both pair strategies need a second instrument. The notebooks use **spot** against the perpetual; `cardinalai/backtest_data` has no spot feed — it is Binance USDⓈ-M perpetuals only. What it does have is the same underlying quoted in two stablecoins: `xrpusdc` and `xrpusdt`. C6 trades the USDC leg and uses the USDT leg as the reference.

Fitted by OLS through the origin on the four pair training sessions:

$$\beta = 0.9913, \qquad R^2 = 0.9985$$

so the pair is about as cointegrated as two instruments get, and both strategies keep their shape. What changes is the economics, and it changes in the direction of a **weaker** result:

- the notebooks' basis is a perpetual's premium over spot — a funding and cash-and-carry object, slow, sign-stable, with a term structure;
- ours is a stablecoin spread plus a venue lead-lag: smaller, faster, and with no funding anchor pulling it back.

The median USDC/USDT basis across the first eight pair sessions runs +0.56 to +8.68 bp, so it is real and sizeable, but it is not the same object. A *better* result here than the notebooks' would be the suspicious one. This is stated before the results because it is exactly the kind of substitution that is easy to report as though it were the published experiment.

5. How wide should the quotes be?

Item 2 of the request — "larger spread (dependent on asset)" — needs the width to come from the asset rather than from taste. Two ways to get it, and the disagreement between them is itself a finding.

5.1 From the tape

`measure_adverse_selection.py` places no orders. It replays ten training days on a 100 ms grid and records, per bucket, the mid and the deepest tick each side's aggressors reached. A quote resting d ticks out is filled in that bucket only if an aggressor reached at least d; an aggressive buy lifts our ask and leaves us short, so the cost is $+(m_{t+h} - m_t)$, and the sign flips on the other side. Volume weighted, because a maker's exposure to a bucket is proportional to how much traded in it.

That gives a cost *curve* over depth rather than C1's single regression. On ETHUSDC at a 10 s horizon, volume weighted, a fill costs 0.82 bp at the touch and rises slowly, to 1.07 bp at 30 ticks and 2.74 bp at 200. What the quote earns — its depth plus the 0.5 bp maker rebate — starts below that and rises faster. They cross at **30 ticks, 0.69 bp**: on the tape, a quote any closer loses and any further out wins.

Weighting matters. Unweighted, the cost at the touch is 0.38 bp rather than 0.82, and the curve sits under the earnings line almost everywhere, which would put the floor at the touch. The big buckets are the expensive ones, and a maker is exposed to them in proportion to their size.

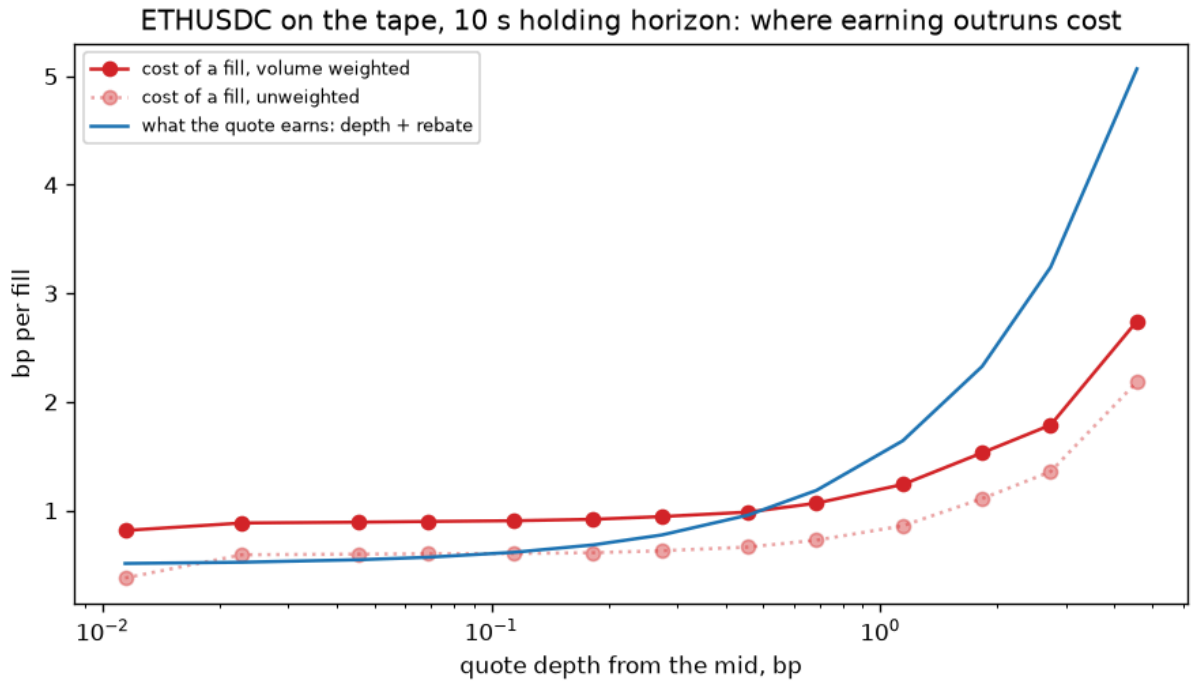


Figure — ETHUSDC, ten training sessions, 10 s holding horizon: what a fill costs on the tape (red), volume weighted and not, against what a quote at that depth earns (blue). Break-even is where blue crosses solid red.

5.2 From the simulator, and why the two disagree

The simulator is far less forgiving. C1's `glft-grid` lost 0.80 bp per fill, and the sweep in §8.1 puts the ETHUSDC floor at 4 bp — six times the tape's 0.69 bp.

The gap is **fill selection**, and the tape cannot see it. A resting quote is not filled by the average trade at its depth; it is filled by the subset of trades that consumed the queue ahead of it, and those are the large informed sweeps. Queue position and latency decide which, and neither is on the tape — they are precisely what hftbacktest's queue model and the synthesised order latency exist to model.

So the tape estimate stays as a diagnostic, and the floor is set by `sweep_spread.py`: run the strategy at each width, on four training days, in the simulator that does model queue position, and read off the best. Four days rather than ten because a sweep is a search and therefore the part of C6 most able to overfit — it is given the least data it can run on, not the most, and the chosen value is then applied unchanged to the held-out set.

6. The RL modification

One environment for all five (`env.py`). The action is four multipliers that every strategy already has a slot for:

action	range	what it scales
`spread_mult`	0.25–8 (0.5–2 forced)	the half spread, however derived

action	range	what it scales
`skew_mult`	0.05–10	the inventory skew
`alpha_mult`	0.05–4	the strategy's own fair-price signal
`grid_mult`	0.25–4	the grid step

mapped logarithmically, because the right prior over a multiplier is uniform in log space. All four are **1.0 at the published parameters**, so each baseline is a strict special case of its own policy — the property C2 built the GLFT environment around, now holding five times over, and checked rather than asserted.

The observation is C2's ten features plus one: the strategy's own signal, in bp. The reward is C2's, unchanged, so C6's numbers are on C3's scale:

$$r_t = \frac{\Delta\,\text{equity}_t}{\text{book}} \times 10^4 - \eta\,\hat{q}_t^2, \qquad \eta = 0.015$$

`alpha_mult` is C5's item in disguise. Setting it to 0 turns a strategy into a pure mid-price quoter with the same spread and skew, so what the policy chooses for it measures what that strategy's signal is worth on these days — and whether the policy leans on the signal more where the signal is informative, which is exactly C5's second item.

TD3 only, two seeds, forced participation. All three are decisions C3 already paid for:

- C3 ran DDPG, TD3 and SAC across three seeds and TD3 won on held-out cumulative return (family medians -2.19 against -4.54 and -4.85). Repeating a three-way comparison five times to re-answer that, on four cores, would buy nothing.
- C3 measured the seed spread on this problem at 3.74 points of return. Two seeds is enough to *notice* that a result is seed-dependent and not enough to rank two close configurations, so any claim here resting on a gap smaller than the seed spread is flagged rather than made.
- C3 established that with a free spread range the highest-scoring policy available on a training day is the corner that quotes so far out it takes no fills and returns exactly zero. Forced participation caps the spread multiplier at 2× so that "do learned parameters beat published ones" is asked of two market makers.

6.1 Recipe v2: what changed after v1 lost to its baseline

The first round (recipe **v1**, everything above) lost to its own widened baseline on median. Three changes were made together as recipe **v2**, and two ablations each remove one of them, so their effects can be told apart.

A residual policy. Every multiplier ranges over $[\frac{1}{2}, 2]$, which is symmetric in log space about 1, so the action 0 *is* the baseline. The actor's last layer is initialised to zero, so training starts exactly on the baseline, and it explores with small Gaussian noise around it instead of v1's first episode of uniformly random parameters. In market making a random episode is not free exploration: it is a day of large, real losses that then dominate the replay buffer. A small

penalty, $0.001\,\text{bp} \times \overline{a^2}$ per step, keeps the policy away from the no-fill corner without pinning it: holding one multiplier at $2^{0.5}$ for a whole session costs about 1 bp, the size of the effects in §8.

Five-second decisions. With a discount of $\gamma = 0.99$ per step, the effective horizon is about $1/(1-\gamma) = 100$ steps. At one-second steps that is 100 seconds, shorter than the minutes over which inventory and adverse selection actually cost money. At five seconds it is about eight minutes.

A counterfactual reward. Replay the same session at the baseline action, record its per-step reward b_t, and train on

$$\tilde{r}_t = r_t - c\, b_t.$$

Because b_t never depends on the policy's action, subtracting it cannot change which policy is optimal. It can only change how noisy the signal pointing to that policy is. The variance of $\tilde{r}$ is

$$\operatorname{Var}(r - c\,b) = \operatorname{Var} r - 2c \operatorname{Cov}(r,b) + c^2 \operatorname{Var} b,$$

minimised at $c^* = \operatorname{Cov}(r,b)/\operatorname{Var} b$. **The first design used $c = 1$, and that was wrong.** On a training session, it *raised* the per-step variance 1.1–2.3× on XRPUSDC. Near the spread floor fills are sparse, so a policy that widens slightly stops trading while the baseline keeps going, and subtracting an uncorrelated series only adds its noise:

probe	corr(r,b)	variance at $c=1$	variance at c^*	c^*
XRP, spread +0.05	0.62	×1.14	×0.62	0.46
XRP, spread +0.10	0.59	×2.31	×0.66	0.31
XRP, exploration noise	0.36	×1.26	×0.87	0.36
ETH, spread +0.05	0.48	×0.93	×0.77	0.54
ETH, spread +0.10	0.55	×0.88	×0.70	0.56
ETH, exploration noise	0.79	×0.43	×0.38	0.77

A fixed $c = 0.5$ recovers most of the optimum in every row and never raises variance (×0.45–0.89 by the formula above). That is what v2 uses. The gain is real but modest, a 10–60% variance cut, and it is reported as that rather than as the order-of-magnitude one might hope for from a control variate.

recipe	step	actions	reward	exploration
v1	1 s	forced ranges	absolute	1 random episode, then noise
v2	5 s	residual $[\frac{1}{2}, 2]$	$r - 0.5\,b$, deviation penalty	noise around the baseline
v2-no-cf	5 s	residual	absolute, deviation penalty	noise around the baseline
v2-no-residual	5 s	forced ranges	$r - 0.5\,b$	1 random episode, then noise

Five seeds. v1 and v2 are both run with five seeds on every (asset, strategy) pair of the two deep assets; the ablations with five seeds on `glft` over XRPUSDC.

7. Checks with known answers

`run_checks.py`, all run before any number below.

check	result
skew unit conversion, C1 ticks vs C6 price	4.95e-16 max relative error over 2,000 random states
OBI running standardisation vs brute-force windowed	2.84e-14 max absolute difference over 7,588 steps
`baseline_action()` maps to multipliers of 1.0	to float32 precision (2e-8), free and forced ranges; exactly under residual ranges
reference tables strictly causal	no row usable before its own timestamp, 500 probes
reward telescopes	to 1e-6 bp over 400 steps
baseline is a strict special case	fills and equity bit-identical to the runner; quoted spread equal to float32 precision
`gymnasium.utils.env_checker`	passes
residual: the action 0 is the baseline	exactly: multipliers $(1, 1, 1, 1)$
residual: zero-initialised actor	outputs exactly 0 for any input, with a non-zero gradient into its last layer
counterfactual reward at the baseline	exactly 0 at every one of a session's 17,160 steps; trace and episode the same length

The last check is the one that rules out a leak. If the cached baseline trace were off by one step, the agent would be handed the *next* step's market move inside its reward, which looks exactly like skill.

Two of the older rows were claimed as exact in the first version of this writeup without having been run: the slow checks had only ever been run with `--skip-slow`. Run, they hold to float32 precision rather than exactly, because the action space is float32. The fills and the equity are bit-identical, so no result moves, but the claim was wrong and the table now says what was measured.

Two of these earned their place by failing first.

The OBI standardisation was wrong. The notebook recomputes `nanmean`/`nanstd` over a trailing window at every step, which is O(window) per step and unaffordable at a 100 ms interval — 36,000 elements at each of 864,000 steps. C6 keeps running sums over a circular buffer instead. The first version divided by the window *occupancy*; the notebook divides by the count of *finite* entries, and on a one-sided book the imbalance is undefined rather than zero. At a 5% hole rate that moved the standardised alpha by up to 0.28 σ. The check caught it before any OBI result existed.

The adverse-selection estimate was reported as a floor, and is not one. Its first version also had the sign backwards, which made adverse selection look like a *gain* — caught because the sign of that number is known in advance. Corrected and volume weighted, it puts the ETHUSDC break-even at 0.69 bp, six times closer than the 4 bp the simulator needs, which is why §5.2 exists and why it does not set the spread.

8. Results

8.1 How wide each asset has to be quoted

The floor is chosen per asset by `sweep_spread.py`, on three or four training sessions, from the widths that still take at least 50 fills a day.

asset	tick, bp	floor, bp	floor / tick
bnbusdc	0.1181	2	17
ethusdc	0.0225	4	178
suiusdc	0.0294	8	273
suiusdt	0.2936	8	27
xrpusdc	0.3428	8	23
xrpusdt	0.3423	8	23
avaxusdc	0.4252	16	38
enausdc	1.5898	32	20
enausdt	1.5911	32	20

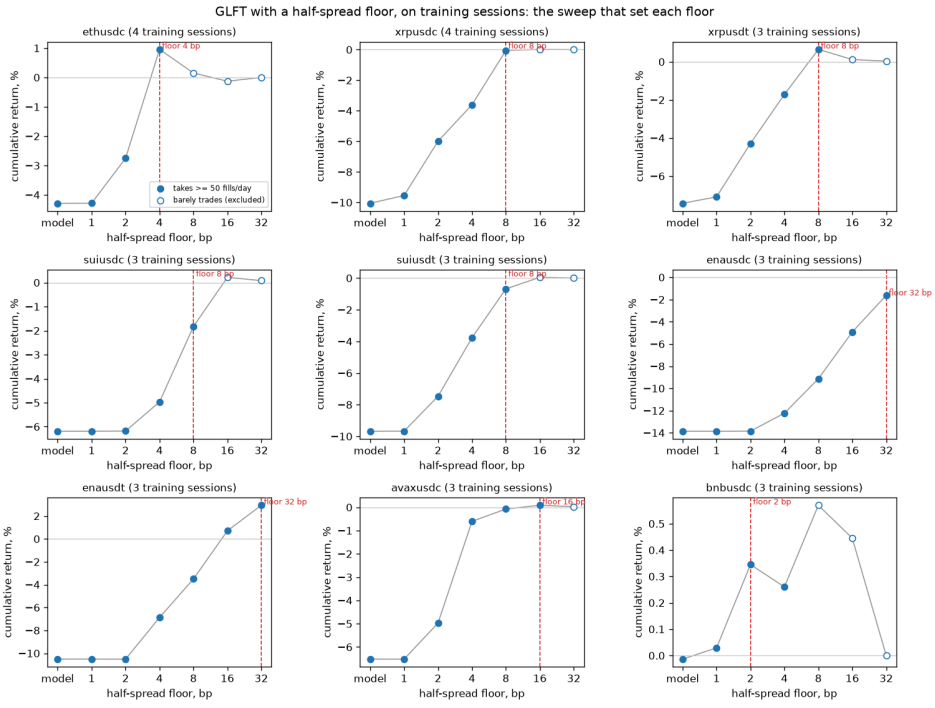


Figure — GLFT with a half-spread floor on each asset's training sessions. Filled points take at least 50 fills a day and are eligible; open points barely trade and are excluded, however well they score. The red line is the floor chosen.

The figure shows two things the table cannot. On `bnbusdc` the best-scoring width (8 bp) is one of the excluded ones: it made +0.57% on a few dozen fills. That is the participation guard doing its job, since a width that barely trades wins by not trading. And on `enausdc` and `enausdt` the curve is still rising at 32 bp, the widest value swept, so their floors mean "at least 32 bp" rather than "32 bp".

Every one of them is wider than what GLFT calibrates to, by between two and twenty times. That is C1's negative result restated across nine assets rather than argued from one.

The floor rises with the tick — log-log correlation 0.756 over the nine — but the claim has to be made carefully, for two reasons. The swept grid is powers of two, so the response variable takes seven values and a correlation on it is coarse. And the ratio is not constant: it sits at 20-38 for the six coarser-tick assets and jumps to 178 and 273 for ETHUSDC and SUIUSDC, the two with the finest ticks. So the tick is not what sets the floor. For a coarse-tick asset the two happen to move together because both are downstream of how much the price moves per unit time; for a fine-tick one there is nothing stopping the quote sitting close, and the binding constraint is adverse selection alone.

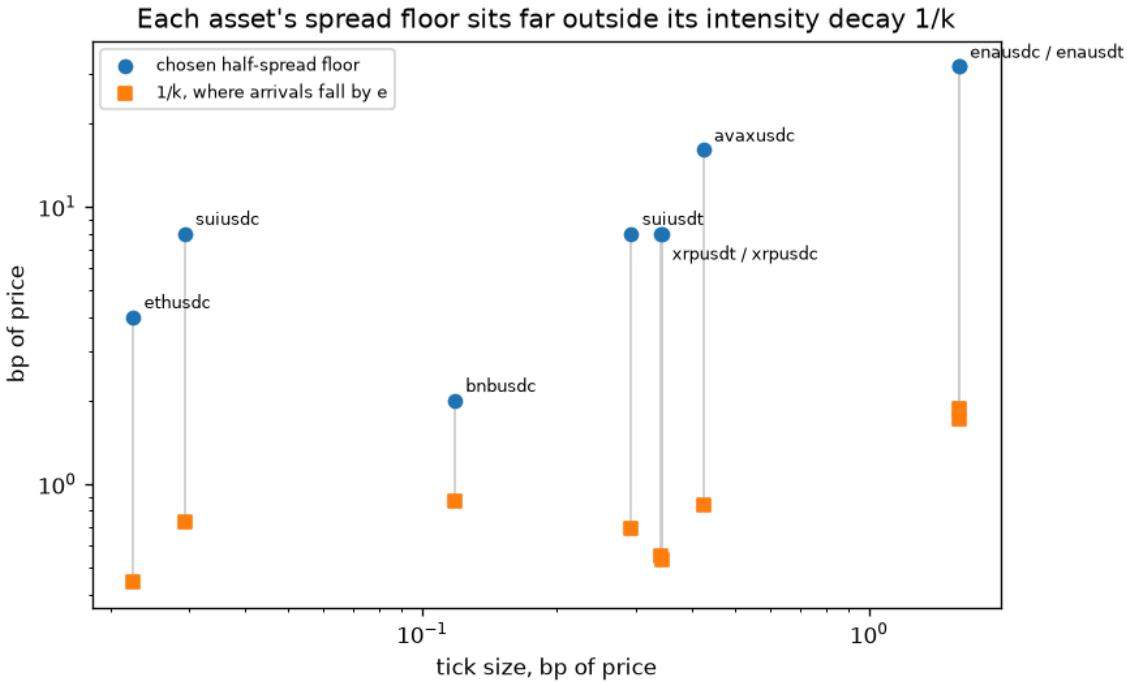


Figure — each asset's chosen floor (blue) and its intensity decay $1/k$ (orange), both in bp, against tick size. The floor sits 9–19× outside $1/k$ on eight of the nine assets (2.3× on `bnbusdc`): the depth at which trades stop arriving is not the depth at which quoting stops losing.

Two entries are a useful internal check rather than new information: `enausdc` and `enausdt` are the same underlying in two stablecoins, and they land on the same floor from independent sweeps, as do `xrpusdc` and `xrpusdt`. `suiusdc` and `suiusdt` also agree at 8 bp despite their ticks differing by a factor of ten, which is the cleanest single piece of evidence that the floor is a property of the asset's flow and not of its tick.

8.2 Widening the spread is worth a lot

Across 33 (asset, strategy) pairs on held-out sessions, replacing each strategy's published half spread with its asset's calibrated floor is worth a **median +2.25 points** of cumulative return, mean +4.32, and helps in **25 of 33** cases. The range is −1.70 to +18.36.

	published	widened
pairs profitable, held out	3 / 33	8 / 33
median cumulative return	−4.59%	−1.74%

That is the single largest effect in Part C. It is also, bluntly, a correction to C1 rather than a discovery: C1 established that the calibrated GLFT spread does not cover adverse selection on ETHUSDC, and the only thing C6 adds is that the same holds on eight more assets and for four more strategies, and that a per-asset floor recovers most of the loss.

Where widening *doesn't* help is as informative. The eight cases where it does not are concentrated in `hfgrid` (four of them) and in the two SUI pairs (four). `hfgrid` already sets its half spread to five times realised volatility, which on these assets is usually wider than the floor -- so the floor rarely binds, and when it does it is removing fills the strategy was profiting from rather than protecting it. `bnbusdc obi` is the clean control: its floor is 2 bp, the strategy already quotes 3.7 bp, the floor never binds, and the published and widened rows are **identical to the digit** (-1.62%, 296.70 fills/h both). A pipeline that can produce that row is at least wired correctly.

8.3 More orders in the grid does nothing

The request asked for deeper grids. Holding order size fixed, so a deeper grid takes no extra risk and only adds price levels, GLFT on XRPUSDC's training sessions gives:

orders per side	published spread: return	fills	widened spread: return	fills
1	−8.85%	46,753	−0.008%	38
5	−9.91%	51,859	−0.014%	170
20	−10.06%	51,925	−0.053%	376
40	−10.05%	51,935	−0.053%	376
60	−10.05%	51,907	−0.053%	376

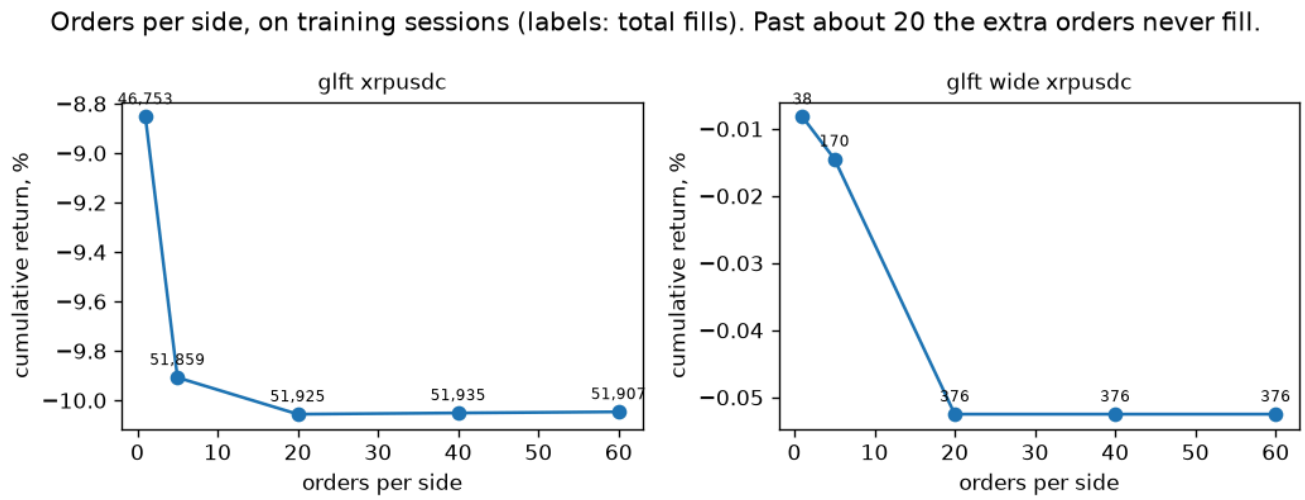


Figure — return against orders per side, labelled with total fills.

Past 20 orders the extra ones never fill. At the widened spread the 20-, 40- and 60-order rows are identical to the digit, because the grid step equals the half spread: the 21st order sits twenty half spreads from the fair price, and nothing reaches it. What people usually mean by "more orders" is finer spacing, and that is the knob that matters — which is why the RL action scales the grid *step* and not the count.

The table also says something less expected: **fewer** orders did better on these sessions, at both spreads. The deep orders only fill during sweeps, and a sweep is exactly when the fill is most adversely selected. That is four training sessions of one asset, so it is a direction to test, not a finding.

8.4 What RL adds on top: v1 never, v2 sometimes

This is the question C6 exists to answer, because C3's version of it was open to the objection that GLFT was a losing baseline. The widened baselines answer that objection. What follows is the answer to the question itself, first with the recipe that C3 used and then with the one built after it lost.

Per-asset equity and inventory paths for every strategy, with the median v1 and v2 seed, are in `docs/assets/c6/` — for example ETHUSDC and XRPUSDC.

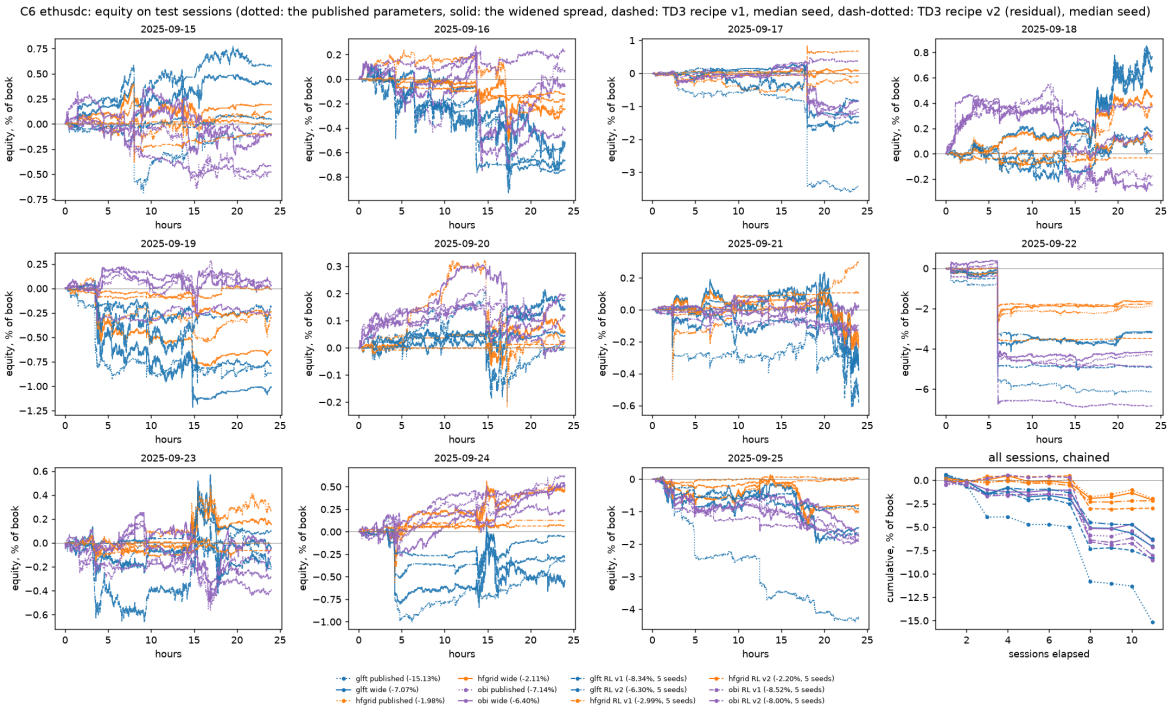


Figure — ETHUSDC, eleven held-out sessions: equity per session and the chained track record. Colour is the strategy; dotted published, solid widened, dashed v1, dash-dotted v2 (the median of five seeds).

The two deep assets, five seeds each. Held-out cumulative return over eleven sessions; each RL cell is the median over five seeds, with the gain over the widened baseline beside it.

asset	strategy	widened	RL v1	v1 gain	RL v2	v2 gain
ETHUSDC	glft	−7.07%	−8.34%	−1.27	−6.30%	**+0.77**

asset	strategy	widened	RL v1	v1 gain	RL v2	v2 gain
ETHUSDC	hfgrid	−2.11%	−2.99%	−0.87	−2.20%	−0.08
ETHUSDC	obi	−6.40%	−8.52%	−2.13	−8.00%	−1.61
XRPUSDC	glft	−1.89%	−2.68%	−0.80	−0.86%	**+1.03**
XRPUSDC	hfgrid	−1.15%	−2.88%	−1.73	−1.48%	−0.33
XRPUSDC	apt	+2.00%	−5.49%	−7.49	−1.66%	−3.66
XRPUSDC	basis	+0.36%	−6.81%	−7.17	−2.35%	−2.71
XRPUSDC	obi	−2.91%	−4.38%	−1.47	−2.15%	**+0.76**

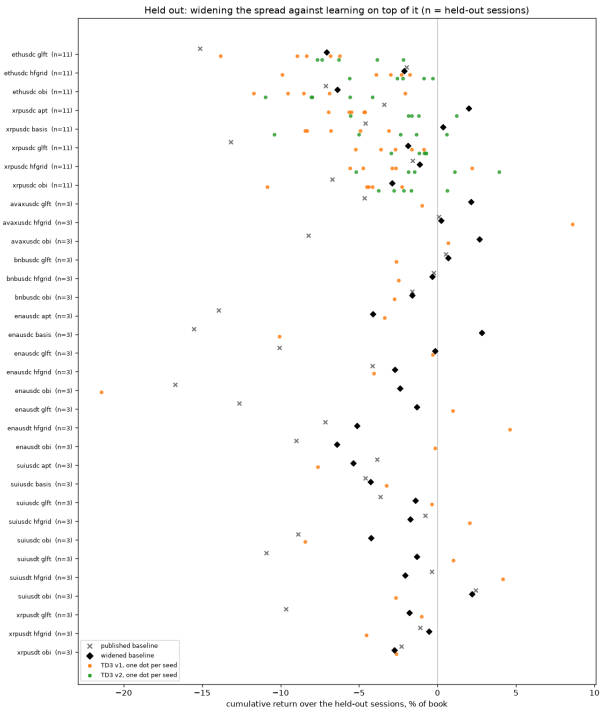
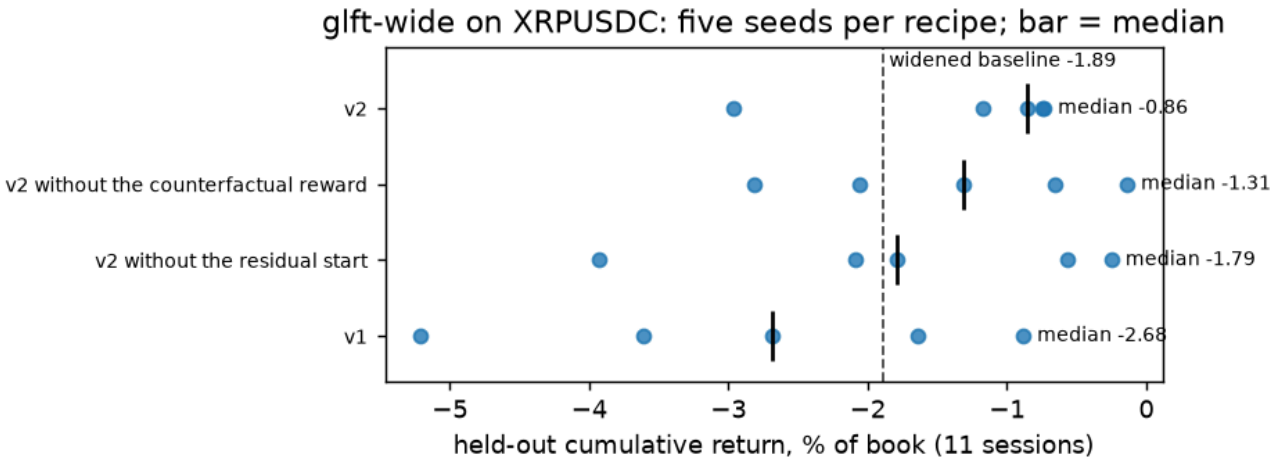


Figure — every (asset, strategy) pair on its held-out sessions: the published baseline (×), the widened baseline (◆), and each trained policy, one dot per seed (v1 orange, v2 green). The breadth assets have one v1 seed each.

Three results:

- **v1 loses to its own baseline in all eight.** Five seeds make that unambiguous: the median v1 policy is between 0.8 and 7.5 points worse than simply quoting the widened spread.
- **v2 beats v1 in all eight**, by a median of +1.9 points, and closes most of the gap to the baseline. It beats the baseline itself in three of eight, by +0.8 to +1.0 points, and loses in five. The median v2 gain is −0.21 points: on these data, v2 is about as good as the baseline it started from, which is what a residual policy that learned little would be.
- **The seed spread is still larger than any gain.** On `glft` ETHUSDC the five v2 seeds run from −7.66% to −2.18%; on `hfgrid` XRPUSDC from −5.20% to +3.91%. A single-seed number would have reported either a large win or a large loss.

Which of v2's changes did the work. The ablations remove one change each, five seeds each, on `glft` over XRPUSDC:



recipe	median	range over seeds
widened baseline	−1.89%	
v1	−2.68%	−5.21 to −0.88
v2 without the residual start	−1.79%	−3.93 to −0.25
v2 without the counterfactual reward	−1.31%	−2.82 to −0.14
v2	**−0.86%**	−2.97 to −0.73

Both help, and they stack. The residual start is worth more: without it the median lands on the baseline (−1.79% against −1.89%), which is what one would expect if the random first episode fills the replay buffer with the losing region and the policy spends the rest of training climbing out. The counterfactual reward adds about another half point and narrows the seed range slightly (2.68 → 2.24 points). Five-second decisions are common to all three v2 variants, so this does not separate their effect from v1's one-second steps.

Over all 33 pairs, which includes the breadth assets with a single v1 seed each:

	pairs	median gain over widened	positive
v1, all pairs	33	−1.27	12 / 33
v1, single-seed pairs	25	−0.14	12 / 25
v1, five-seed pairs	8	−1.60	0 / 8
v2, five-seed pairs	8	−0.21	3 / 8

Single-seed rows are kinder to RL than five-seed rows, which is what one expects if a single seed is partly a draw from the spread rather than a measurement of it. The five-seed rows are the honest estimate.

By strategy, the median points v1 adds to its own widened baseline:

strategy	n	median v1 gain
glft	9	−0.14
hfgrid	9	−0.87
obi	9	−1.99

strategy	n	median v1 gain
apt	3	−2.25
basis	3	−7.17

`glft` is again the one where RL comes closest, and it is also the only one whose half spread is already a live function of market state — `A`, `k` and σ, refit every five seconds — rather than a constant. The pair strategies are the worst. Their alpha term is the only thing separating them from a plain grid, so a plausible reading is that a policy scaling it mostly scales noise; the probe in §8.5 is consistent with that, not proof of it.

So the C3 result survives the obvious objection, and a better recipe narrows it without reversing it. Learned, state-conditioned multipliers are at best as good as well-chosen fixed ones here — on five strategies and nine assets, with five seeds where it counts. C7 then asks what information the policy was missing, and finds that a three-line rule conditioned on the markouts of our own fills beats every recipe here on each of the six pairs it was run on.

8.5 Does the policy respond to the alpha signal? (C5's second item)

C5 asks whether RL discovers the *interaction* between an alpha signal and the quoting parameters on its own, or merely reproduces what a person would write by hand -- and says plainly that a negative answer is a finding.

The answer is no. Replaying a held-out session and correlating what each policy saw against what it emitted, over the six policies whose signal actually moved during the session (`sd(signal) >= 0.25`):

	median |r|	sign agreement
signal → alpha weight	0.104	3 of 6 positive
signal → skew	0.082	2 of 6 positive
inventory → skew (control)	**0.121**	—

The response to the alpha signal is *weaker than the response to inventory*, and its sign is a coin flip across assets within the same strategy. There is no interaction here to have discovered.

Two caveats, both of which cut the same way. The control is itself small: 0.121 is a weak response to the inventory penalty these policies were explicitly trained against, so the honest reading is "these policies are barely state-conditioned on anything" rather than "they learned everything except the signal". And two further policies were dropped from the median because their signal barely varied (`sd` 0.088 and 0.016) -- a correlation against a near-constant is not a measurement, which is why `sd(signal)` is a reported column rather than a filter applied silently.

So for this problem the analytic model with a hand-written alpha term is the better engineering choice, which is exactly the outcome C5 anticipated and asked to have checked rather than assumed.

How not to answer this. The first attempt swept the signal feature through a trained policy with the other features held neutral and read off the response. It gives large, confident-looking numbers -- and its negative control destroys them. `glft` quotes off the mid, so its signal feature is identically zero throughout training and its policy cannot have learned anything about that input; swept the same way, `glft` policies move the alpha weight across 181% of its range against 181% for policies whose signal is real. Sweeping an input a network never saw vary measures extrapolation. Both scripts are kept (`probe_policy.py`, `probe_in_session.py`) because the failing control is the more useful of the two results.

Raw output: `c6-probe.txt`.

9. Where these items come from

what	source	why this one
the request	user message following C3	five attached notebooks, four explicit asks
`glft`	GLFT Market Making Model and Grid Trading	C1's baseline; the phase's point of comparison
`hfgrid`	High-Frequency Grid Trading	the same grid without any intensity model — isolates what the GLFT fit buys
`apt`	Market Making with Alpha — APT	a lead-lag fair price; the only one of the five whose signal is exogenous
`basis`	Market Making with Alpha — Basis	a carry signal, and the one with a real economic anchor
`obi`	Market Making with Alpha — Order Book Imbalance	endogenous signal, highest published Sharpe of the five
GLFT equations (4.6)/(4.7)	Guéant, Lehalle & Fernandez-Tapia	the closed form the whole of Part C is built on
the widened split	`config.py`	supersets of C1's lists; re-picking would invalidate C3
TD3, seeds, forced participation	C3	each was settled there; C6 spends the budget on breadth
the reward and the ten base features	C2	unchanged, so C6's numbers sit on C3's scale
`alpha_mult` as C5's item	`ladder.md`	C5 asks for the alpha signals in the state and for whether the policy responds to them